Creating and Working on the Sales database

```sql
CREATE DATABASE SalesDB;
USE SalesDB;
```

Creating the tables :

Customers and Regions:

```sql
CREATE TABLE Customers (
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(100) NOT NULL UNIQUE
);

CREATE TABLE Regions (
    RegionID INT PRIMARY KEY,
    RegionName VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50) NOT NULL UNIQUE
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
36	15:31:14	SELECT s.StudentID, s.Name FROM Students s WHERE s.StudentID IN (SELECT e.StudentID FROM ...	5 row(s) returned	0.015 sec / 0.000 sec
37	15:37:09	CREATE DATABASE SalesDB	1 row(s) affected	0.016 sec
38	15:37:09	USE SalesDB	0 row(s) affected	0.000 sec
39	15:38:18	CREATE TABLE Customers (CustomerID INT PRIMARY KEY, CustomerName VARCHAR(100) NOT NU...	0 row(s) affected	0.062 sec
40	15:38:18	CREATE TABLE Regions (RegionID INT PRIMARY KEY, RegionName VARCHAR(50) NOT NULL UNI...	0 row(s) affected	0.047 sec
41	15:38:18	CREATE TABLE Products (ProductID INT PRIMARY KEY, ProductName VARCHAR(50) NOT NULL U...	0 row(s) affected	0.047 sec

Orders Table :

```sql
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    CustomerID INT NOT NULL,
    RegionID INT NOT NULL,
    OrderDate DATE NOT NULL,

    CONSTRAINT fk_orders_customer
        FOREIGN KEY (CustomerID)
        REFERENCES Customers(CustomerID),

    CONSTRAINT fk_orders_region
        FOREIGN KEY (RegionID)
        REFERENCES Regions(RegionID)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
37	15:37:09	CREATE DATABASE SalesDB	1 row(s) affected	0.016 sec
38	15:37:09	USE SalesDB	0 row(s) affected	0.000 sec
39	15:38:18	CREATE TABLE Customers (CustomerID INT PRIMARY KEY, CustomerName VARCHAR(100) NOT NU...	0 row(s) affected	0.062 sec
40	15:38:18	CREATE TABLE Regions (RegionID INT PRIMARY KEY, RegionName VARCHAR(50) NOT NULL UNI...	0 row(s) affected	0.047 sec
41	15:38:18	CREATE TABLE Products (ProductID INT PRIMARY KEY, ProductName VARCHAR(50) NOT NULL U...	0 row(s) affected	0.047 sec
42	15:39:04	CREATE TABLE Orders (Order_ID INT PRIMARY KEY, CustomerID INT NOT NULL, RegionID INT N...	0 row(s) affected	0.046 sec

OrderItems

```
37 • CREATE TABLE OrderItems (
38     OrderItemID INT PRIMARY KEY,
39     Order_ID INT NOT NULL,
40     ProductID INT NOT NULL,
41     Quantity INT NOT NULL,
42     UnitPrice DECIMAL(12,2) NOT NULL,
43
44     CONSTRAINT chk_quantity
45         CHECK (Quantity > 0),
46
47     CONSTRAINT chk_unit_price
48         CHECK (UnitPrice >= 0),
49
50     CONSTRAINT fk_items_order
51         FOREIGN KEY (Order_ID)
52         REFERENCES Orders(Order_ID),
53
54     CONSTRAINT fk_items_product
55         FOREIGN KEY (ProductID)
56         REFERENCES Products(ProductID),
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
38	15:37:09	USE SalesDB	0 row(s) affected	0.000 sec
39	15:38:18	CREATE TABLE Customers (CustomerID INT PRIMARY KEY, CustomerName VARCHAR(100) NOT NU...	0 row(s) affected	0.062 sec
40	15:38:18	CREATE TABLE Regions (RegionID INT PRIMARY KEY, RegionName VARCHAR(50) NOT NULL UNI...	0 row(s) affected	0.047 sec
41	15:38:18	CREATE TABLE Products (ProductID INT PRIMARY KEY, ProductName VARCHAR(50) NOT NULL U...	0 row(s) affected	0.047 sec
42	15:39:04	CREATE TABLE Orders (Order_ID INT PRIMARY KEY, CustomerID INT NOT NULL, RegionID INT N...	0 row(s) affected	0.046 sec
43	15:40:17	CREATE TABLE OrderItems (OrderItemID INT PRIMARY KEY, Order_ID INT NOT NULL, ProductID I...	0 row(s) affected	0.078 sec

Inserting the Data :

```
62 • INSERT INTO Customers (CustomerID, CustomerName)
63   VALUES
64   (1, 'Rahul Sharma'),
65   (2, 'Priya Reddy'),
66   (3, 'Amit Kumar'),
67   (4, 'Sneha Rao'),
68   (5, 'Ananya Reddy'),
69   (6, 'Meena Devi'),
70   (7, 'Naveen Kumar'),
71   (8, 'Divya Sharma');
72
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
39	15:38:18	CREATE TABLE Customers (CustomerID INT PRIMARY KEY, CustomerName VARCHAR(100) NOT NU...	0 row(s) affected	0.062 sec
40	15:38:18	CREATE TABLE Regions (RegionID INT PRIMARY KEY, RegionName VARCHAR(50) NOT NULL UNI...	0 row(s) affected	0.047 sec
41	15:38:18	CREATE TABLE Products (ProductID INT PRIMARY KEY, ProductName VARCHAR(50) NOT NULL U...	0 row(s) affected	0.047 sec
42	15:39:04	CREATE TABLE Orders (Order_ID INT PRIMARY KEY, CustomerID INT NOT NULL, RegionID INT N...	0 row(s) affected	0.046 sec
43	15:40:17	CREATE TABLE OrderItems (OrderItemID INT PRIMARY KEY, Order_ID INT NOT NULL, ProductID I...	0 row(s) affected	0.078 sec
44	15:41:14	INSERT INTO Customers (CustomerID, CustomerName) VALUES (1, 'Rahul Sharma'), (2, 'Priya Reddy'), (3, 'A...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.016 sec

```
74 • INSERT INTO Regions (RegionID, RegionName)
75   VALUES
76   (1, 'Hyderabad'),
77   (2, 'Khammam'),
78   (3, 'Warangal');
79
80 • INSERT INTO Products (ProductID, ProductName)
81   VALUES
82   (1, 'Laptop'),
83   (2, 'Mobile'),
84   (3, 'Headphones'),
85   (4, 'Tablet');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
41	15:38:18	CREATE TABLE Products (ProductID INT PRIMARY KEY, ProductName VARCHAR(50) NOT NULL U...	0 row(s) affected	0.047 sec
42	15:39:04	CREATE TABLE Orders (Order_ID INT PRIMARY KEY, CustomerID INT NOT NULL, RegionID INT N...	0 row(s) affected	0.046 sec
43	15:40:17	CREATE TABLE OrderItems (OrderItemID INT PRIMARY KEY, Order_ID INT NOT NULL, ProductID I...	0 row(s) affected	0.078 sec
44	15:41:14	INSERT INTO Customers (CustomerID, CustomerName) VALUES (1, 'Rahul Sharma'), (2, 'Priya Reddy'), (3, 'A...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.016 sec
45	15:41:52	INSERT INTO Regions (RegionID, RegionName) VALUES (1, 'Hyderabad'), (2, 'Khammam'), (3, 'Warangal')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
46	15:41:52	INSERT INTO Products (ProductID, ProductName) VALUES (1, 'Laptop'), (2, 'Mobile'), (3, 'Headphones'), (4, '...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec

```sql
INSERT INTO Orders (Order_ID, CustomerID, RegionID, OrderDate)
VALUES
(1001, 1, 1, '2026-01-05'),
(1002, 2, 2, '2026-01-06'),
(1003, 3, 3, '2026-01-07'),
(1004, 4, 1, '2026-01-08'),
(1007, 5, 1, '2026-01-11'),
(1010, 6, 1, '2026-01-14'),
(1014, 2, 2, '2026-01-18'),
(1015, 7, 2, '2026-01-19'),
(1016, 8, 3, '2026-01-20');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
42	15:39:04	CREATE TABLE Orders (Order_ID INT PRIMARY KEY, CustomerID INT NOT NULL, RegionID INT N...	0 row(s) affected	0.046 sec
43	15:40:17	CREATE TABLE OrderItems (OrderItemID INT PRIMARY KEY, Order_ID INT NOT NULL, ProductID I...	0 row(s) affected	0.078 sec
44	15:41:14	INSERT INTO Customers (CustomerID, CustomerName) VALUES (1, 'Rahul Sharma'), (2, 'Priya Reddy'), (3, 'A...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.016 sec
45	15:41:52	INSERT INTO Regions (RegionID, RegionName) VALUES (1, 'Hyderabad'), (2, 'Khammam'), (3, 'Warangal')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
46	15:41:52	INSERT INTO Products (ProductID, ProductName) VALUES (1, 'Laptop'), (2, 'Mobile'), (3, 'Headphones'), (4, '...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
47	15:42:48	INSERT INTO Orders (Order_ID, CustomerID, RegionID, OrderDate) VALUES (1001, 1, 1, '2026-01-05'), (1002...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0.016 sec

```sql
INSERT INTO OrderItems
    (OrderItemID, Order_ID, ProductID, Quantity, UnitPrice)
VALUES
(1, 1001, 1, 2, 55000),
(2, 1002, 2, 3, 18000),
(3, 1003, 3, 5, 1500),
(4, 1004, 4, 1, 22000),
(5, 1007, 3, 2, 1500),
(6, 1010, 1, 2, 55000),
(7, 1014, 4, 2, 22000),
(8, 1015, 1, 3, 55000),
(9, 1016, 2, 5, 1558000);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
43	15:40:17	CREATE TABLE OrderItems (OrderItemID INT PRIMARY KEY, Order_ID INT NOT NULL, ProductID I...	0 row(s) affected	0.078 sec
44	15:41:14	INSERT INTO Customers (CustomerID, CustomerName) VALUES (1, 'Rahul Sharma'), (2, 'Priya Reddy'), (3, 'A...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.016 sec
45	15:41:52	INSERT INTO Regions (RegionID, RegionName) VALUES (1, 'Hyderabad'), (2, 'Khammam'), (3, 'Warangal')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
46	15:41:52	INSERT INTO Products (ProductID, ProductName) VALUES (1, 'Laptop'), (2, 'Mobile'), (3, 'Headphones'), (4, '...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
47	15:42:48	INSERT INTO Orders (Order_ID, CustomerID, RegionID, OrderDate) VALUES (1001, 1, 1, '2026-01-05'), (1002...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0.016 sec
48	15:44:22	INSERT INTO OrderItems (OrderItemID, Order_ID, ProductID, Quantity, UnitPrice) VALUES (1, 1001, 1, 2, 5...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0.015 sec

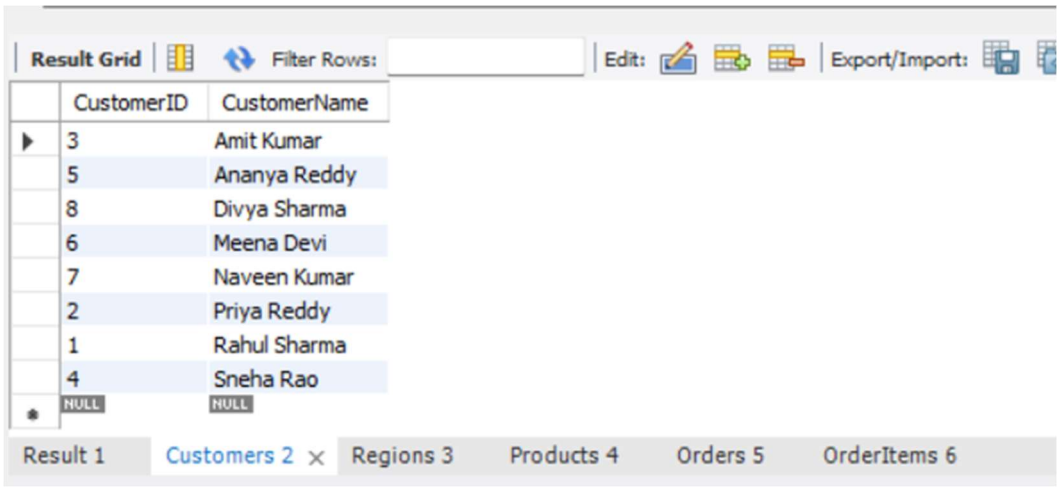
Checking

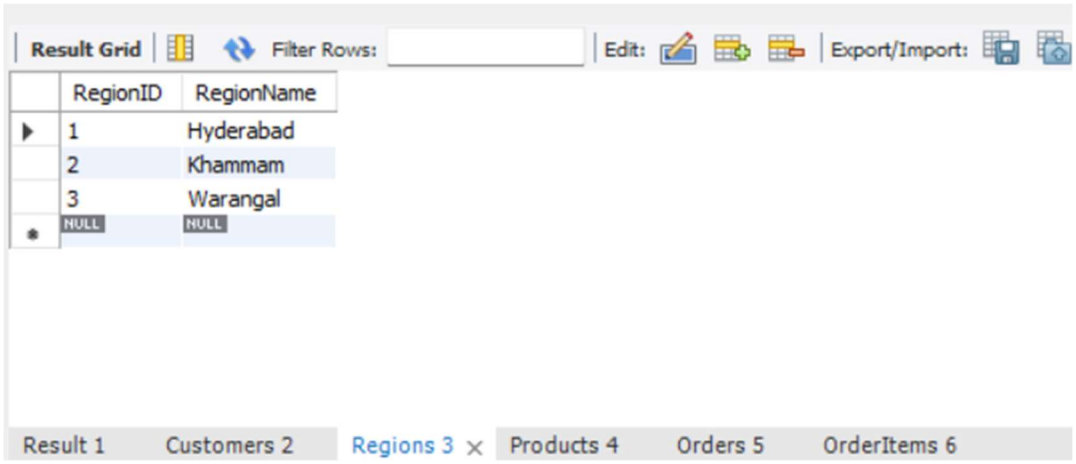
```sql
SHOW TABLES;

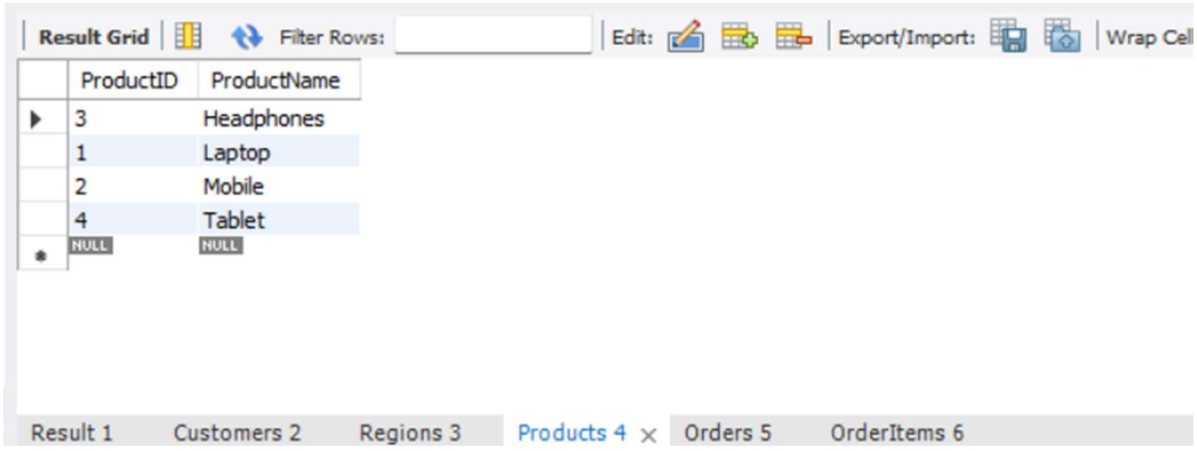
SELECT * FROM Customers;
SELECT * FROM Regions;
SELECT * FROM Products;
SELECT * FROM Orders;
SELECT * FROM OrderItems;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Tables_in_salesdb
customers
orderitems
orders
products
regions


Result Grid
Filter Rows:
Edit:
Export/Import:
CustomerID
CustomerName
3
Amit Kumar
5
Ananya Reddy
8
Divya Sharma
6
Meena Devi
7
Naveen Kumar
2
Priya Reddy
1
Rahul Sharma
4
Sneha Rao
NULL
NULL
Result 1
Customers 2
Regions 3
Products 4
Orders 5
OrderItems 6


Result Grid
Filter Rows:
Edit:
Export/Import:
RegionID
RegionName
1
Hyderabad
2
Khammam
3
Warangal
NULL
NULL
Result 1
Customers 2
Regions 3
Products 4
Orders 5
OrderItems 6


Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell
ProductID
ProductName
3
Headphones
1
Laptop
2
Mobile
4
Tablet
NULL
NULL
Result 1
Customers 2
Regions 3
Products 4
Orders 5
OrderItems 6

Result Grid | Filter Rows: | Edit: | Export/Import:

Order_ID	CustomerID	RegionID	OrderDate
1001	1	1	2026-01-05
1002	2	2	2026-01-06
1003	3	3	2026-01-07
1004	4	1	2026-01-08
1007	5	1	2026-01-11
1010	6	1	2026-01-14
1014	2	2	2026-01-18
1015	7	2	2026-01-19
1016	8	3	2026-01-20

Result 1 | Customers 2 | Regions 3 | Products 4 | Orders 5 | OrderItems 6

Result Grid | Filter Rows: | Edit: | Export/Import:

OrderItemID	Order_ID	ProductID	Quantity	UnitPrice
1	1001	1	2	55000.00
2	1002	2	3	18000.00
3	1003	3	5	1500.00
4	1004	4	1	22000.00
5	1007	3	2	1500.00
6	1010	1	2	55000.00
7	1014	4	2	22000.00
8	1015	1	3	55000.00
9	1016	2	5	1558000.00

Result 1 | Customers 2 | Regions 3 | Products 4 | Orders 5 | OrderItems 6

Queries :

1. Find orders placed after 10 January 2026.

```
121 •  SELECT * FROM Orders WHERE OrderDate > '2026-01-10';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Order_ID	CustomerID	RegionID	OrderDate
1007	5	1	2026-01-11
1010	6	1	2026-01-14
1014	2	2	2026-01-18
1015	7	2	2026-01-19
1016	8	3	2026-01-20
NULL	NULL	NULL	NULL

2. Find orders with a quantity greater than two.

```
123 •  SELECT * FROM OrderItems WHERE Quantity > 2;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

OrderItemID	Order_ID	ProductID	Quantity	UnitPrice
2	1002	2	3	18000.00
3	1003	3	5	1500.00
8	1015	1	3	55000.00
9	1016	2	5	1558000.00
NULL	NULL	NULL	NULL	NULL

3. Find products priced between ₹10,000 and ₹60,000.

```
126 •  SELECT *
127    FROM Products p
128    JOIN OrderItems oi ON p.ProductID = oi.ProductID
129    WHERE oi.UnitPrice BETWEEN 10000 AND 60000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ProductID	ProductName	OrderItemID	Order_ID	ProductID	Quantity	UnitPrice
1	Laptop	1	1001	1	2	55000.00
2	Mobile	2	1002	2	3	18000.00
4	Tablet	4	1004	4	1	22000.00
1	Laptop	6	1010	1	2	55000.00
4	Tablet	7	1014	4	2	22000.00
1	Laptop	8	1015	1	3	55000.00

4. Find orders from Hyderabad or Warangal.

```
131 •  SELECT o.* FROM Orders o
132    JOIN Regions r ON o.RegionID = r.RegionID
133    WHERE r.RegionName IN ('Hyderabad', 'Warangal');
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_ID	CustomerID	RegionID	OrderDate
1001	1	1	2026-01-05
1004	4	1	2026-01-08
1007	5	1	2026-01-11
1010	6	1	2026-01-14
1003	3	3	2026-01-07
1016	8	3	2026-01-20

Using aggregate functions

5. Calculate revenue for every order item

```
136 •  SELECT Order_ID,Quantity,UnitPrice,Quantity * UnitPrice AS Revenue FROM OrderItems;
137
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_ID	Quantity	UnitPrice	Revenue
1001	2	55000.00	110000.00
1002	3	18000.00	54000.00
1003	5	1500.00	7500.00
1004	1	22000.00	22000.00
1007	2	1500.00	3000.00
1010	2	55000.00	110000.00
1014	2	22000.00	44000.00
1015	3	55000.00	165000.00
1016	5	1558000.00	7790000.00

6. Calculate total revenue, average order-item value, highest price and lowest price.

```
139 •  SELECT
140        SUM(Quantity * UnitPrice) AS TotalRevenue,
141        ROUND(AVG(Quantity * UnitPrice), 2) AS AvgItemRevenue,
142        MAX(UnitPrice) AS HighestUnitPrice,
143        MIN(UnitPrice) AS LowestUnitPrice
144    FROM OrderItems;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TotalRevenue	AvgItemRevenue	HighestUnitPrice	LowestUnitPrice
8305500.00	922833.33	1558000.00	1500.00

JOIN queries

7.INNER JOIN — orders with customer names

```
147 •  SELECT o.Order_ID,c.CustomerName,o.OrderDate
148    FROM Orders o
149    INNER JOIN Customers c
150        ON o.CustomerID = c.CustomerID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_ID	CustomerName	OrderDate
1003	Amit Kumar	2026-01-07
1007	Ananya Reddy	2026-01-11
1016	Divya Sharma	2026-01-20
1010	Meena Devi	2026-01-14
1015	Naveen Kumar	2026-01-19
1002	Priya Reddy	2026-01-06
1014	Priya Reddy	2026-01-18
1001	Rahul Sharma	2026-01-05

8. Calculate revenue by region

```
166 •  SELECT
167        r.RegionName,
168        COUNT(DISTINCT o.Order_ID) AS TotalOrders,
169        SUM(oi.Quantity * oi.UnitPrice) AS TotalRevenue
170    FROM Regions r
171    JOIN Orders o
172        ON r.RegionID = o.RegionID
173    JOIN OrderItems oi
174        ON o.Order_ID = oi.Order_ID
175    GROUP BY r.RegionID, r.RegionName
176    ORDER BY TotalRevenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

RegionName	TotalOrders	TotalRevenue
Warangal	2	7797500.00
Khammam	3	263000.00
Hyderabad	4	245000.00

Subqueries

9. Find order items priced above the average unit price

```
178 •  SELECT Order_ID, ProductID, UnitPrice
179    FROM OrderItems
180    WHERE UnitPrice > (
181        SELECT AVG(UnitPrice)
182        FROM OrderItems
183    );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

Order_ID	ProductID	UnitPrice
1016	2	1558000.00

10. Categorize order values

```
186 •  SELECT
187        Order_ID,
188        Quantity * UnitPrice AS Revenue,
189        CASE
190            WHEN Quantity * UnitPrice >= 100000 THEN 'High Value'
191            WHEN Quantity * UnitPrice >= 30000 THEN 'Medium Value'
192            ELSE 'Low Value'
193        END AS OrderCategory
194    FROM OrderItems;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_ID	Revenue	OrderCategory
1001	110000.00	High Value
1002	54000.00	Medium Value
1003	7500.00	Low Value
1004	22000.00	Low Value
1007	3000.00	Low Value
1010	110000.00	High Value
1014	44000.00	Medium Value